31338 Network Servers
Lab 7a: Managing filesystems, including mounting and unmounting

Aims
1. To explore the filesystems of existing machines
2. To understand the purpose of the setgid bit, and how to share files among groups in UNIX
3. To add a new filesystem to `/etc/fstab`, and to mount and unmount it
4. To mount and unmount removable media

Task 1: Gathering information on the host OS and your virtual machine

On the host operating system of a Linux workstation in the Faculty labs, determine and document the structure of the filesystem. Include all of the currently mounted filesystems. How many hard disks are in the machine? Do they appear to be IDE or SCSI disks? How many partitions are there on the disk(s)? Identify which partitions are primary, extended and logical partitions. Which are logical volumes? Where are home directories stored? Where is the `/tmp` directory stored?

Hints: Look at both `/proc/partitions` and the fstab file. You can also look in `/dev/disk/by-uuid` or use the `/sbin/blkid` command to figure out the mapping of UUIDs to partitions (blkid may require root privileges).

Repeat this process for your virtual machine OS. How many (virtual) hard disks? Can you tell from the device file names whether the disks are SATA, IDE or SCSI? How many partitions? This time use the `fdisk -l` command to print the partition table as well as looking at `/proc/partitions` and the fstab file. Which partitions are primary, extended and logical? Where are home directories stored? Where is the `/tmp` directory stored? This time you can also refer to the previous lab on disk partitioning for other ways to view disk partitions (as opposed to mounted filesystems). Because this requires root privileges, you couldn't do it on the host OS.

Task 2: Using file permissions to support file sharing among users

In an earlier lab exercise, you should have created four users named Peter, Stewie, Brian and Lois, and assigned two of those users, Stewie and Brian, to a group called *family*. This task builds on that earlier work. Let us assume that Stewie and Brian have the groups stewie and brian as their primary groups respectively (the default case under CentOS, Fedora and Red Hat Linux), but both are also members of the family group. That is, the family group is a secondary group for Stewie and Brian. If this is not the case, you might like to adjust group memberships to comply with the above. This makes the task more difficult, but more interesting.

Now assume that members of the *family* group need to share files with each other. Make a directory called `/share/family`. Make it so that Stewie and Brian (members of the *family* group) can read and create files in this directory, but Peter (who is not a member of the family group) cannot. Make sure that in this directory, files created by Stewie can be read and edited by Brian and vice versa (but they should not be world readable).

To solve this problem, you will need to use three things:

- UNIX file permissions on the `/share/family` directory, including the SetGID bit (`chmod` command)
- Group ownership of the `/share/family` directory (`chgrp` command)
- The `umask` settings of Stewie and Brian, so that files are created group writable by default (`umask` command, e.g. in their `.bashrc file`)

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Task 3: Making /tmp a separate filesystem, and testing mounting and unmounting

In your virtual machine, change the `/tmp` directory from being part of the root filesystem to using a `tmpfs` filesystem. First, create an empty file in the `/tmp` directory (e.g. "`touch /tmp/mytest`"). Then make the necessary change to the `/etc/fstab` file so that `/tmp` uses `tmpfs`, as follows:

```
tmpfs        /tmp         tmpfs        defaults     0 0
```

Then mount the filesystem with '`mount /tmp`'. Use the `mount` command again (with no arguments) to show currently mounted filesystems and verify that `/tmp` is now using `tmpfs`. Also verify that the file you created earlier (`/tmp/mytest`) is not there any more in the `tmpfs` filesystem (where is it?).

After doing this, you might want to change `/tmp` back to the way it was before, so as not to degrade the performance of your virtual machine by using `tmpfs` (since your VM does not use very much memory in total).

Task 4: Mounting removable media (optional)

This lab may not work on the UTS lab machines, but will work on your own laptop or at home. It is optional.

If you have a CD-ROM or a USB flash drive, you can set it up so that it is mounted in your virtual machine. Examples of mount commands to mount these kind of media might be:

```
mount –t iso9660 /dev/cdrom /media/cdrom
mount –t vfat /dev/sdb1 /media/usbdisk
```

On your virtual machine, you will need to create the mount points `/media/cdrom` and/or `/media/usbdisk`. Also when mounting a USB device, the location could be `/dev/sdb1` or may be `/dev/sdc1`, etc, depending on what other devices are in the system.

Your ask is to now add entries to the `/etc/fstab` file for these devices. They should not be mounted when the machine boots though (use the option "`noauto`" in the fstab file).

